Faculty of Computing (FC)

Riphah International University, Islamabad, Pakistan

Semester: V

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SAP ID: 59717

Instructor: Mr. Junaid khan

Course: Data Warehouse & BI

Department: BS Data Science

Section: DS 5-1

Lab Task 5:

Code:

```
CREATE DATABASE LibraryManagement;

GO

USE LibraryManagement;

GO

CREATE TABLE Authors (AuthorID INT PRIMARY KEY IDENTITY(1,1), AuthorName VARCHAR(100) NOT NULL);

CREATE TABLE Publishers ( PublisherID INT PRIMARY KEY IDENTITY(1,1), PublisherName VARCHAR(100) NOT NULL UNIQUE);

CREATE TABLE Books ( BookID INT PRIMARY KEY IDENTITY(1,1), Title VARCHAR(150) NOT NULL, ISBN VARCHAR(20) UNIQUE,
PublisherID INT, PublishYear INT,  CopiesAvailable INT NOT NULL DEFAULT 1,  FOREIGN KEY (PublisherID) REFERENCES
Publishers(PublisherID),  CHECK (CopiesAvailable >= 0));

CREATE TABLE BookAuthors (

   BookID INT,

   AuthorID INT,

   PRIMARY KEY (BookID, AuthorID), FOREIGN KEY (BookID) REFERENCES Books(BookID),  FOREIGN KEY (AuthorID) REFERENCES
Authors(AuthorID));

CREATE TABLE Members (  MemberID INT PRIMARY KEY IDENTITY(1,1), MemberName VARCHAR(100) NOT NULL, Email
VARCHAR(100) UNIQUE, Phone VARCHAR(20), JoinDate DATE DEFAULT GETDATE() );

CREATE TABLE Librarians ( LibrarianID INT PRIMARY KEY IDENTITY(1,1), LibrarianName VARCHAR(100) NOT NULL, Email
VARCHAR(100) UNIQUE);

CREATE TABLE BorrowRecords (

   BorrowID INT PRIMARY KEY IDENTITY(1,1),

   BookID INT NOT NULL,

   MemberID INT NOT NULL,

   LibrarianID INT NOT NULL,

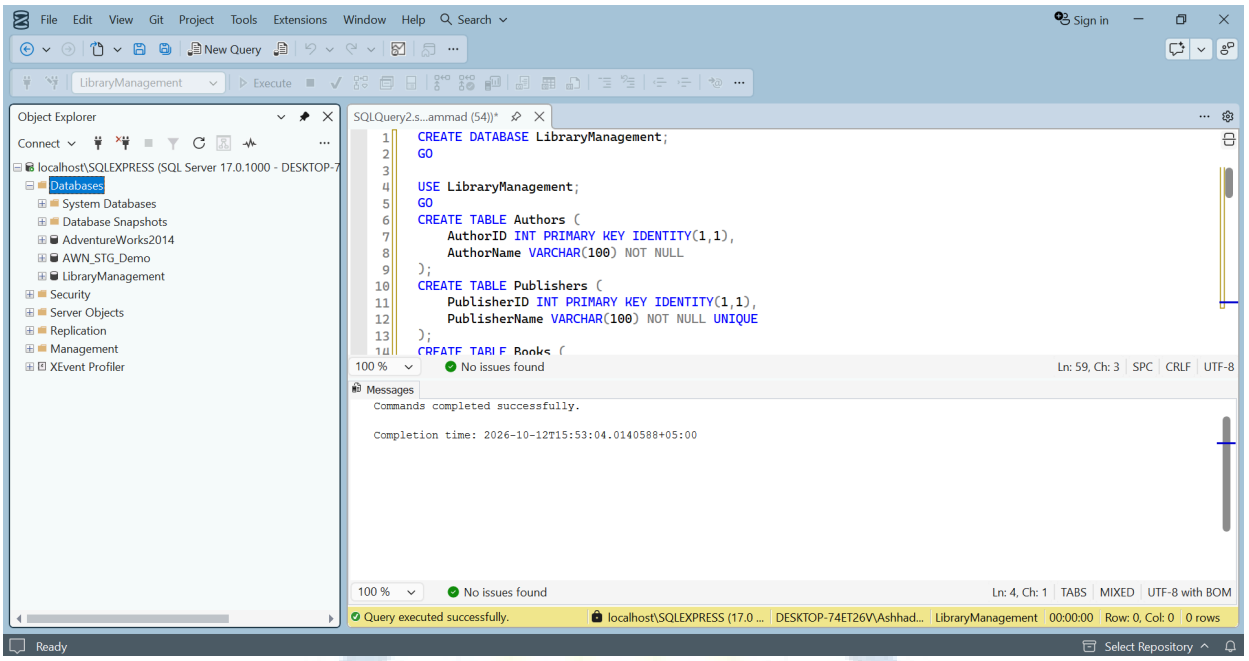
   BorrowDate DATE DEFAULT GETDATE(),

   DueDate DATE NOT NULL,

   ReturnDate DATE NULL,

   FOREIGN KEY (BookID) REFERENCES Books(BookID), FOREIGN KEY (MemberID) REFERENCES Members(MemberID),
FOREIGN KEY (LibrarianID) REFERENCES Librarians(LibrarianID),   CHECK (DueDate >= BorrowDate),   CHECK (ReturnDate IS NULL
OR ReturnDate >= BorrowDate));

GO
```

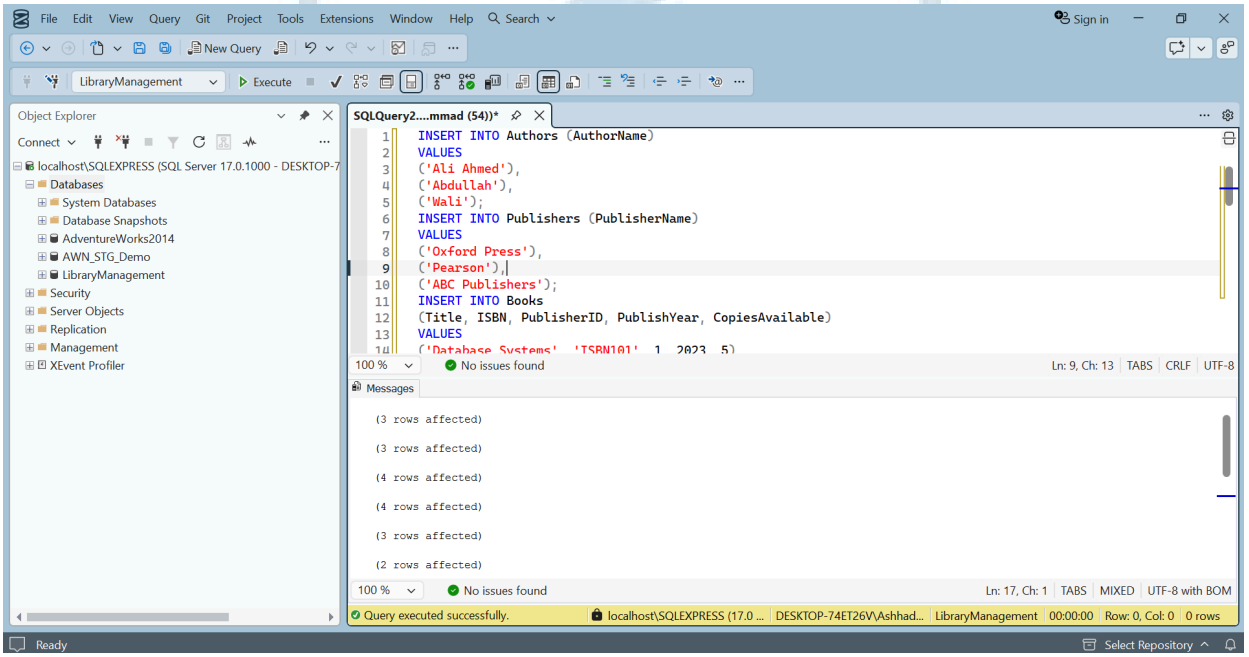


Code:

```
INSERT INTO Authors (AuthorName)
VALUES
('Ali Ahmed'),
('Abdullah'),
('Wali');
INSERT INTO Publishers (PublisherName)
VALUES
('Oxford Press'),
('Pearson'),
('ABC Publishers');
INSERT INTO Books
(Title, ISBN, PublisherID, PublishYear, CopiesAvailable)
VALUES
('Database Systems', 'ISBN101', 1, 2023, 5),
('Data Science Basics', 'ISBN102', 2, 2024, 3),
('SQL for Beginners', 'ISBN103', 3, 2022, 4),
('Artificial Intelligence', 'ISBN104', 1, 2025, 2);
INSERT INTO BookAuthors (BookID, AuthorID)
VALUES
(1, 1),
(2, 2),
(3, 1),
```

(4, 3);

```
INSERT INTO Members (MemberName, Email, Phone)
VALUES
('Ahmed Khan', 'ahmed@gmail.com', '03001234567'),
('Usman Ali', 'usman@gmail.com', '03111234567'),
('Sara Ahmed', 'sara@gmail.com', '03221234567');
INSERT INTO Librarians (LibrarianName, Email)
VALUES
('Mr. Hassan', 'hassan@library.com'),
('Ms. Ayesha', 'ayesha@library.com');
INSERT INTO BorrowRecords
(BookID, MemberID, LibrarianID, BorrowDate, DueDate, ReturnDate)
VALUES
(1, 1, 1, '2026-10-01', '2026-10-15', NULL),
(2, 2, 1, '2026-10-02', '2026-10-16', '2026-10-10'),
(3, 1, 2, '2026-10-03', '2026-10-17', NULL),
(4, 3, 2, '2026-10-04', '2026-10-18', NULL);
GO
```



Code:

```
SELECT * FROM Authors;

SELECT * FROM Publishers;

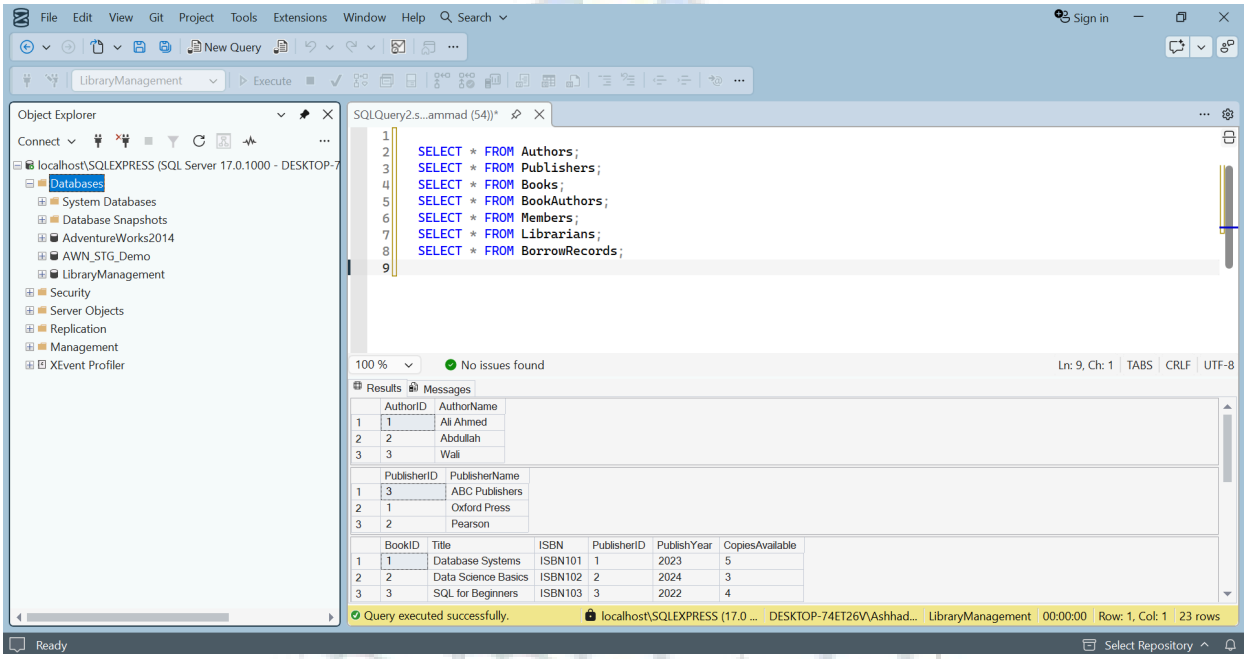
SELECT * FROM Books;

SELECT * FROM BookAuthors;

SELECT * FROM Members;

SELECT * FROM Librarians;

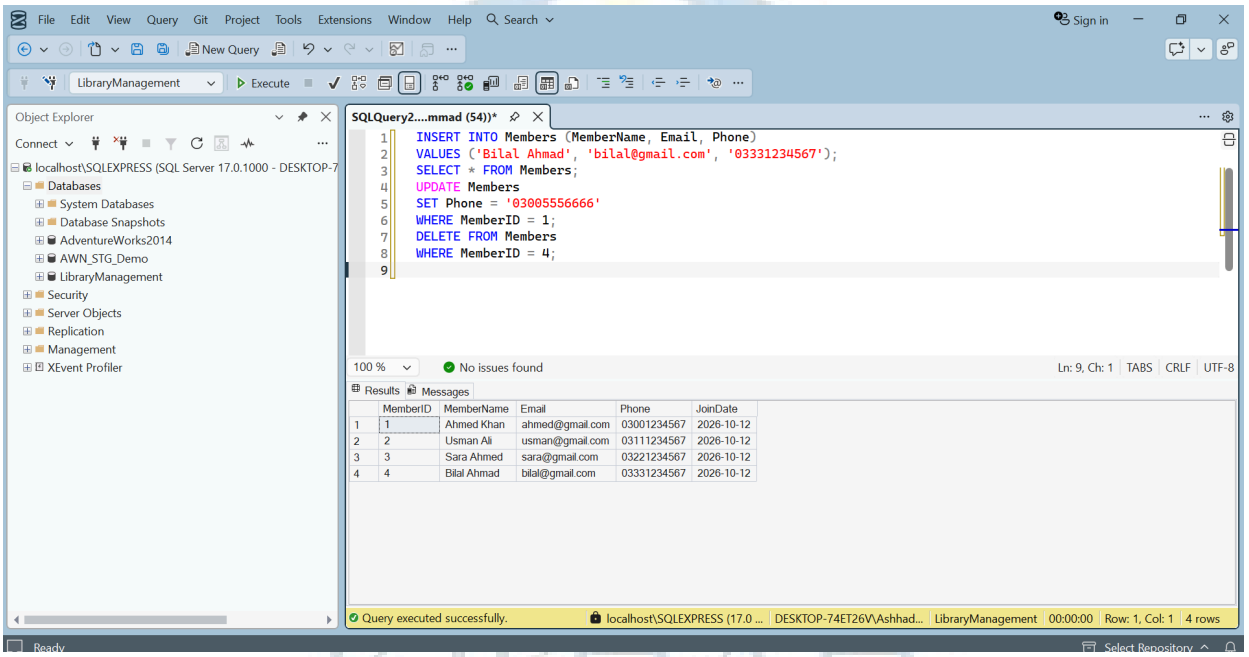
SELECT * FROM BorrowRecords;
```



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Code:

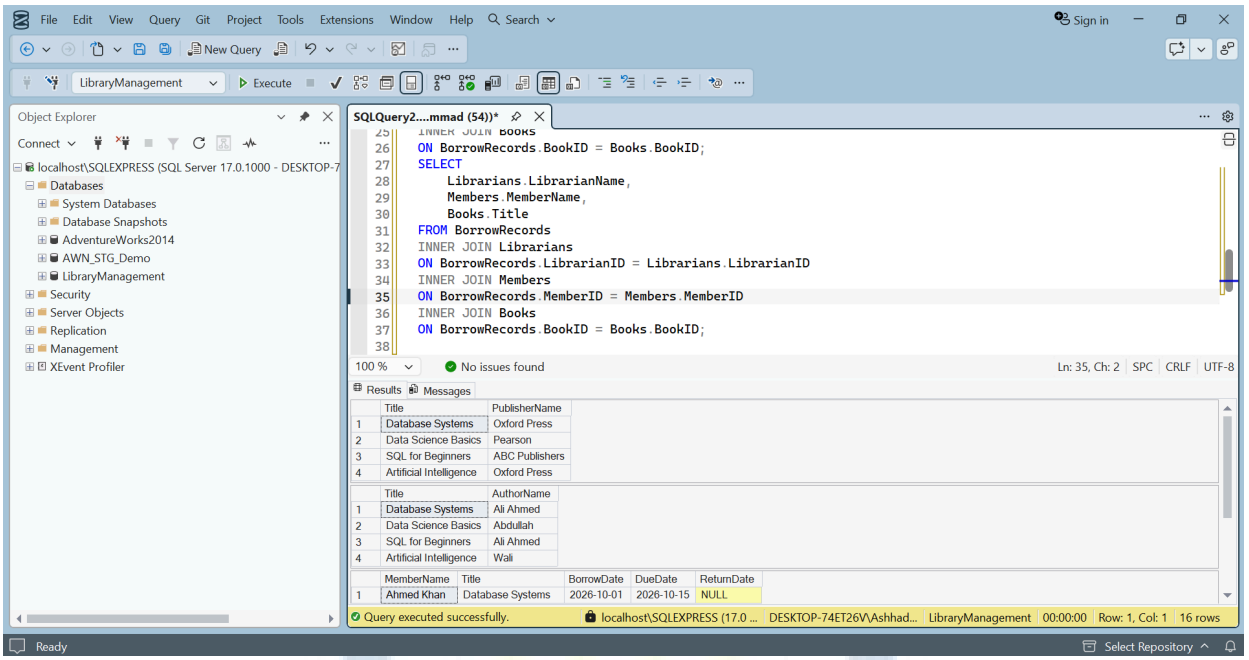
```
INSERT INTO Members (MemberName, Email, Phone)
VALUES ('Bilal Ahmad', 'bilal@gmail.com', '03331234567');
SELECT * FROM Members;
UPDATE Members
SET Phone = '03005556666'
WHERE MemberID = 1;
DELETE FROM Members
WHERE MemberID = 4;
```



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Code:

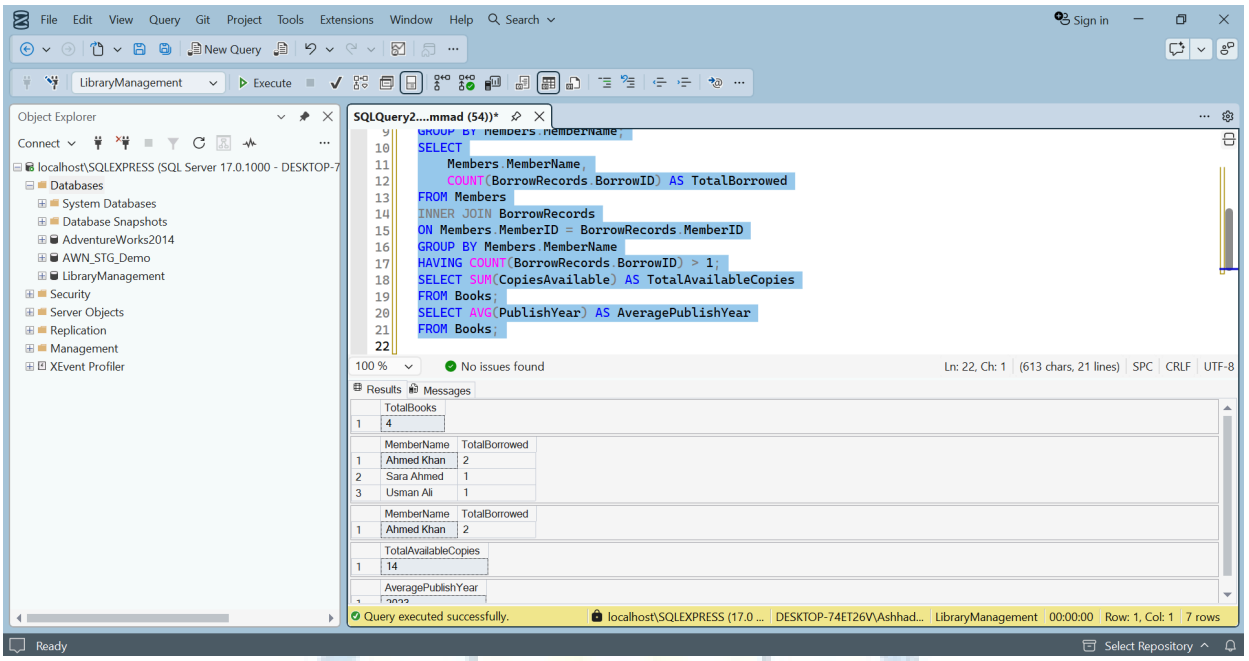
```
SELECT
    Books.Title,
    Publishers.PublisherName
FROM Books
INNER JOIN Publishers
ON Books.PublisherID = Publishers.PublisherID;
SELECT
    Books.Title,
    Authors.AuthorName
FROM Books
INNER JOIN BookAuthors
ON Books.BookID = BookAuthors.BookID
INNER JOIN Authors
ON BookAuthors.AuthorID = Authors.AuthorID;
SELECT
    Members.MemberName,
    Books.Title,
    BorrowRecords.BorrowDate,
    BorrowRecords.DueDate,
    BorrowRecords.ReturnDate
FROM BorrowRecords
INNER JOIN Members
ON BorrowRecords.MemberID = Members.MemberID
INNER JOIN Books
ON BorrowRecords.BookID = Books.BookID;
SELECT
    Librarians.LibrarianName,
    Members.MemberName,
    Books.Title
FROM BorrowRecords
INNER JOIN Librarians
ON BorrowRecords.LibrarianID = Librarians.LibrarianID
INNER JOIN Members
ON BorrowRecords.MemberID = Members.MemberID
INNER JOIN Books
ON BorrowRecords.BookID = Books.BookID;
```



Code:

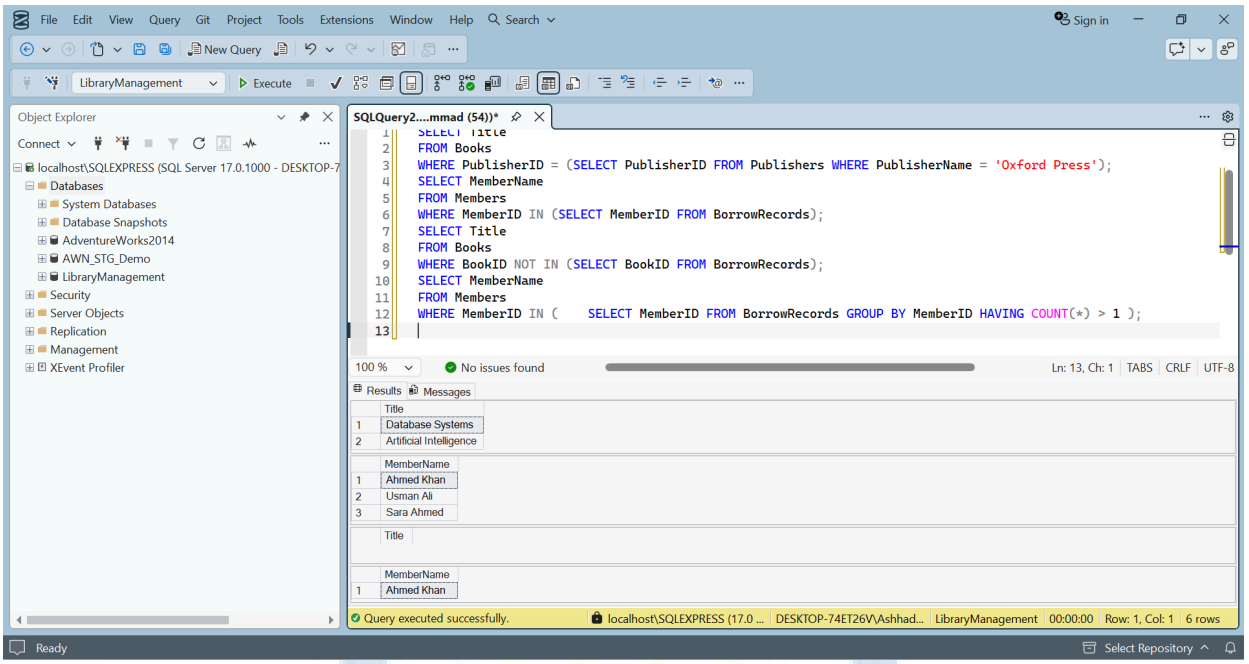
```
SELECT COUNT(*) AS TotalBooks
FROM Books;
SELECT
   Members.MemberName,
   COUNT(BorrowRecords.BorrowID) AS TotalBorrowed
FROM Members
LEFT JOIN BorrowRecords
ON Members.MemberID = BorrowRecords.MemberID
GROUP BY Members.MemberName;
SELECT
   Members.MemberName,
   COUNT(BorrowRecords.BorrowID) AS TotalBorrowed
FROM Members
INNER JOIN BorrowRecords
ON Members.MemberID = BorrowRecords.MemberID
GROUP BY Members.MemberName
HAVING COUNT(BorrowRecords.BorrowID) > 1;
SELECT SUM(CopiesAvailable) AS TotalAvailableCopies
FROM Books;
SELECT AVG(PublishYear) AS AveragePublishYear
```

FROM Books;



Code:

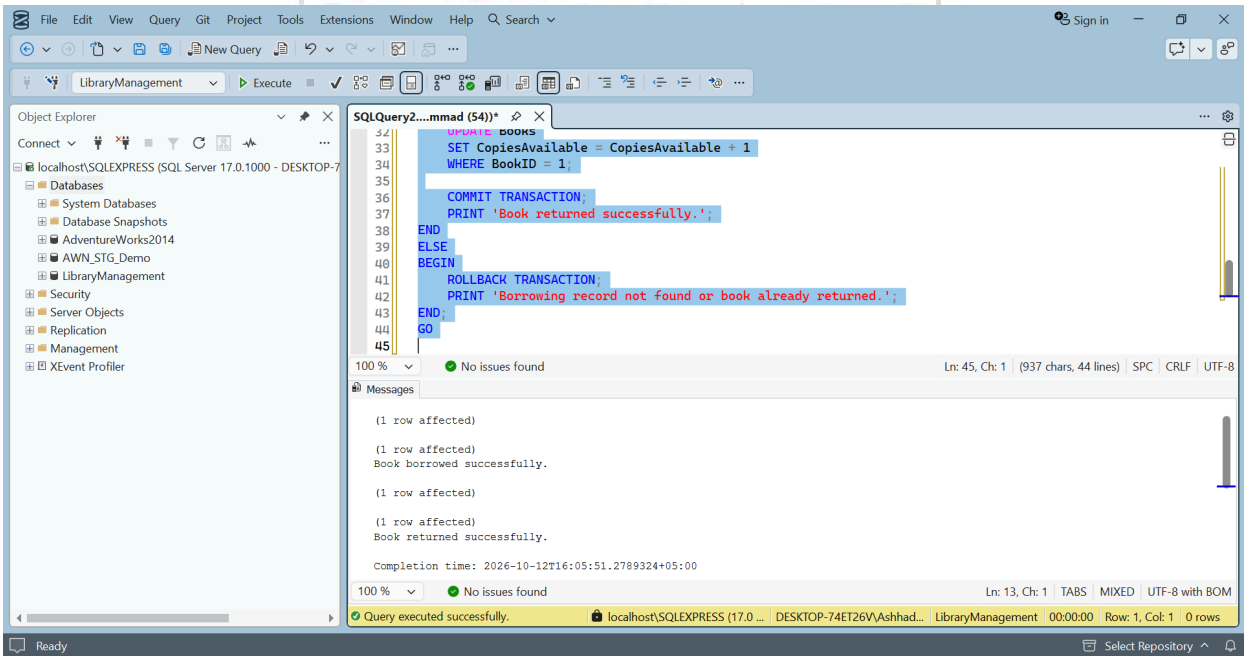
```
SELECT Title
FROM Books
WHERE PublisherID = (SELECT PublisherID FROM Publishers WHERE PublisherName = 'Oxford Press');
SELECT MemberName
FROM Members
WHERE MemberID IN (SELECT MemberID FROM BorrowRecords);
SELECT Title
FROM Books
WHERE BookID NOT IN (SELECT BookID FROM BorrowRecords);
SELECT MemberName
FROM Members
WHERE MemberID IN (    SELECT MemberID FROM BorrowRecords GROUP BY MemberID HAVING COUNT(*) > 1 );
```



Code:

```
BEGIN TRANSACTION;
UPDATE Books
SET CopiesAvailable = CopiesAvailable - 1
WHERE BookID = 1
AND CopiesAvailable > 0;
IF @@ROWCOUNT = 1
BEGIN
  INSERT INTO BorrowRecords
  (BookID, MemberID, LibrarianID, BorrowDate, DueDate)
  VALUES
  (1, 1, 1, CAST(GETDATE() AS DATE),
   DATEADD(DAY, 14, CAST(GETDATE() AS DATE)));
  COMMIT TRANSACTION;
  PRINT 'Book borrowed successfully.';
END
ELSE
BEGIN
  ROLLBACK TRANSACTION;
```

```
    PRINT 'Book is not available.';
END;
GO
BEGIN TRANSACTION;
UPDATE BorrowRecords
SET ReturnDate = CAST(GETDATE() AS DATE)
WHERE BorrowID = 1
AND ReturnDate IS NULL;
IF @@ROWCOUNT = 1
BEGIN
    UPDATE Books
    SET CopiesAvailable = CopiesAvailable + 1
    WHERE BookID = 1;
    COMMIT TRANSACTION;
    PRINT 'Book returned successfully.';
END
ELSE
BEGIN
    ROLLBACK TRANSACTION;
    PRINT 'Borrowing record not found or book already returned.';
END;
GO
```



Code:

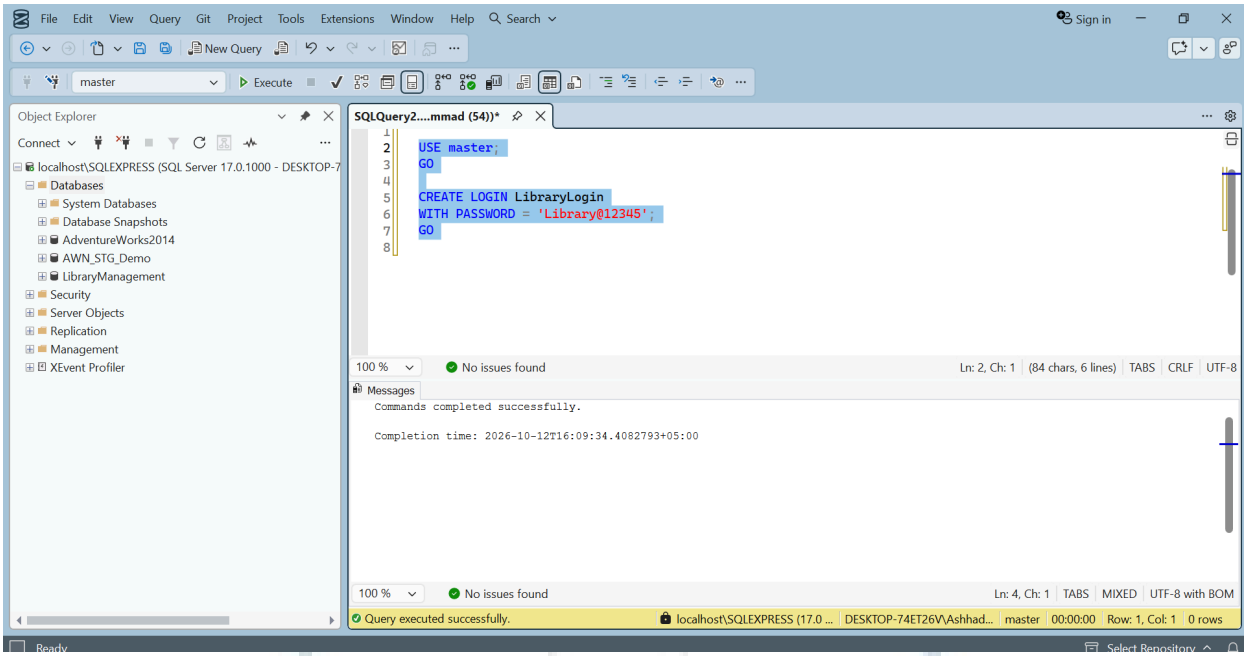
```
USE master;

GO

CREATE LOGIN LibraryLogin

WITH PASSWORD = 'Library@12345';

GO
```



Code:

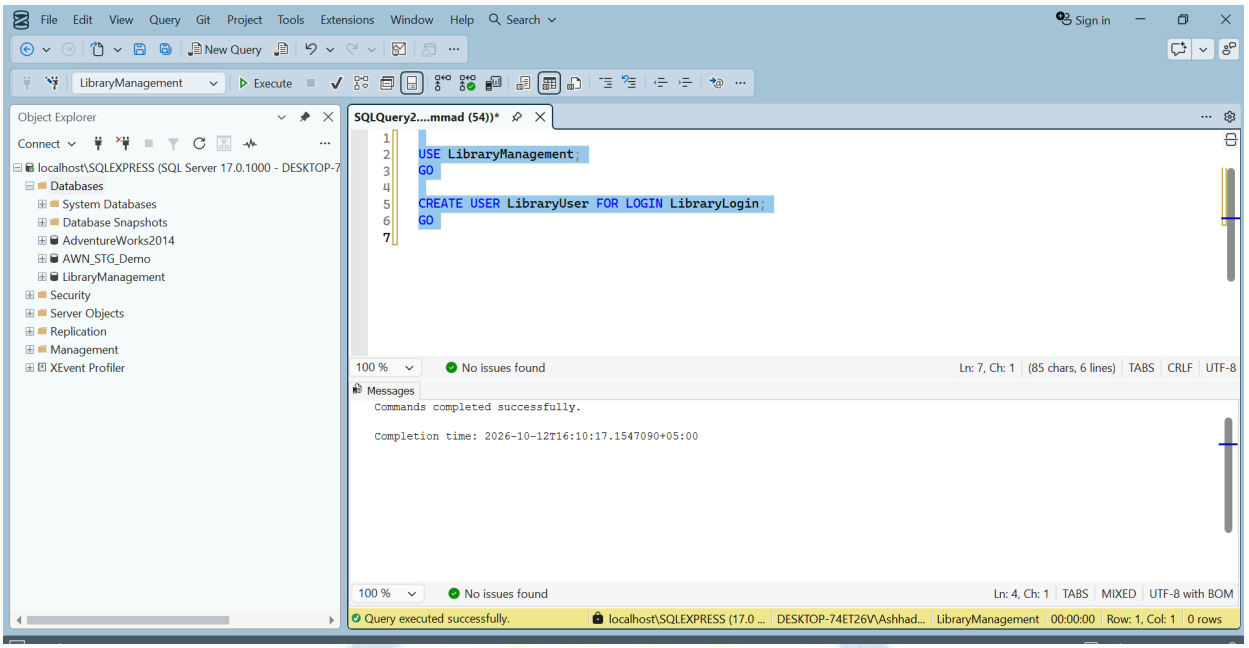
```
USE LibraryManagement;

GO

CREATE USER LibraryUser FOR LOGIN LibraryLogin;

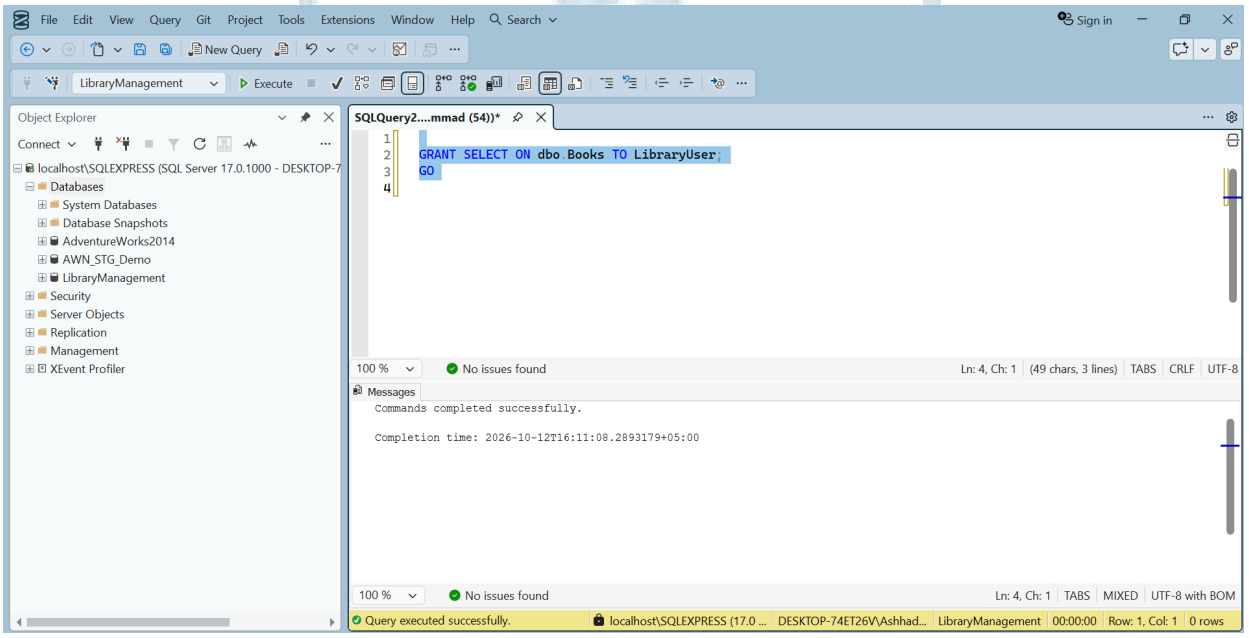
GO
```

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Code:

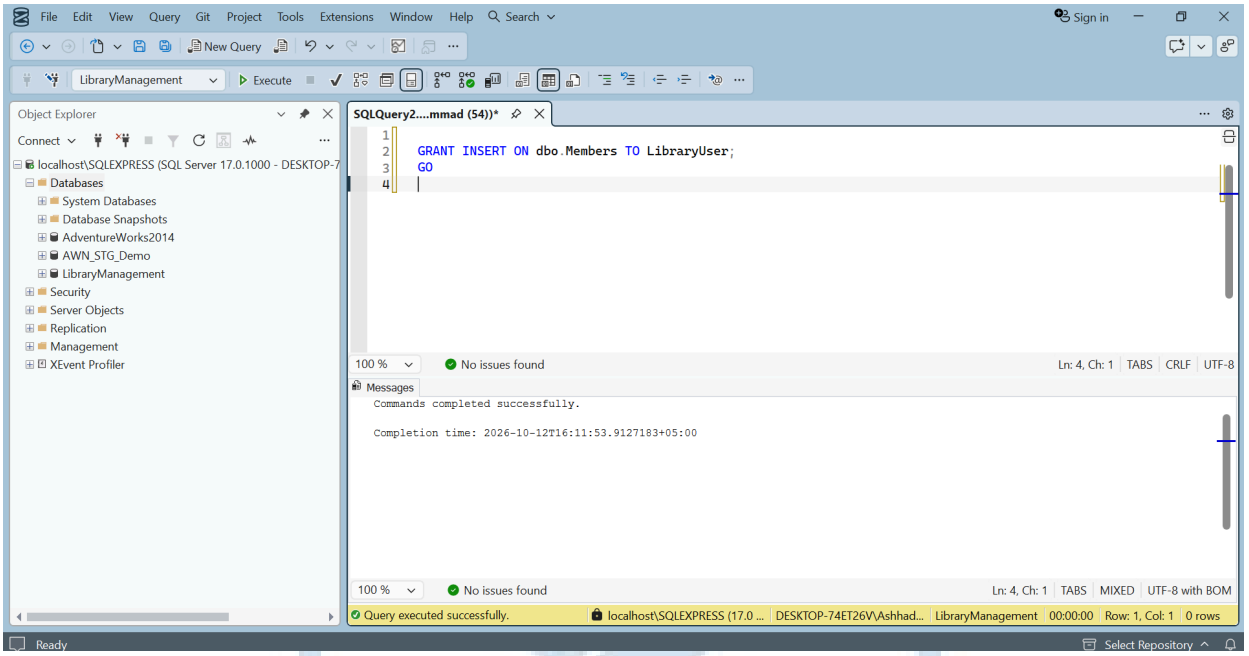
```
GRANT SELECT ON dbo.Books TO LibraryUser;
GO
```



Code:

```
GRANT INSERT ON dbo.Members TO LibraryUser;

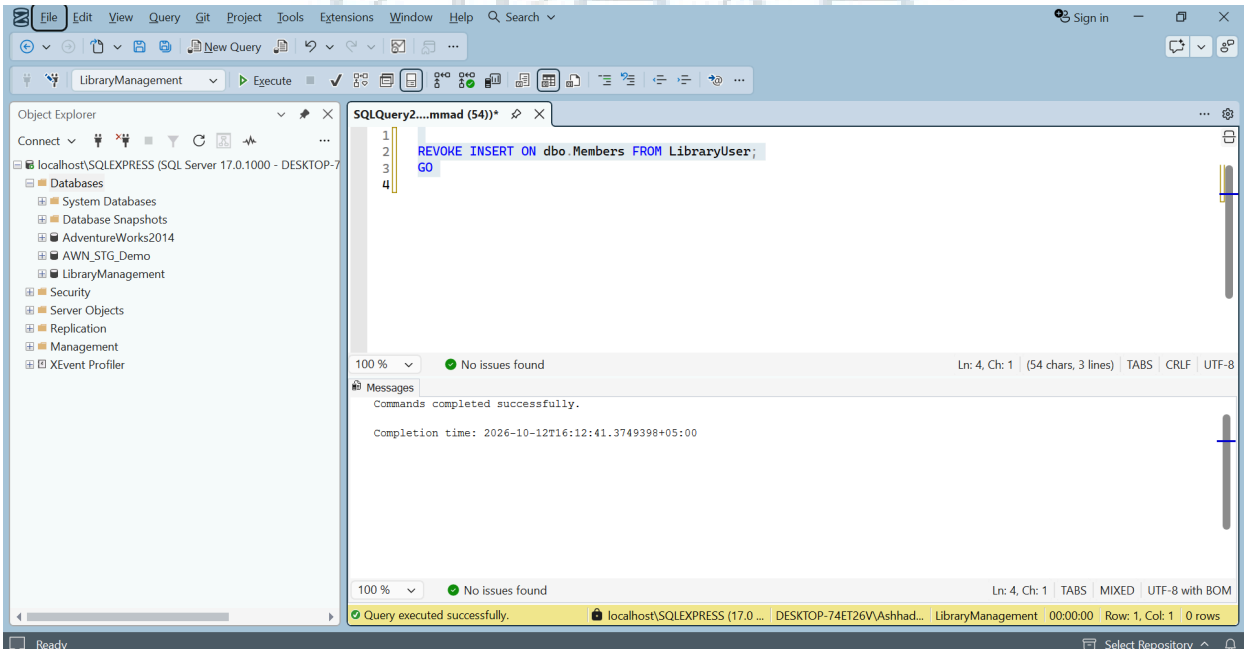
GO
```



Code:

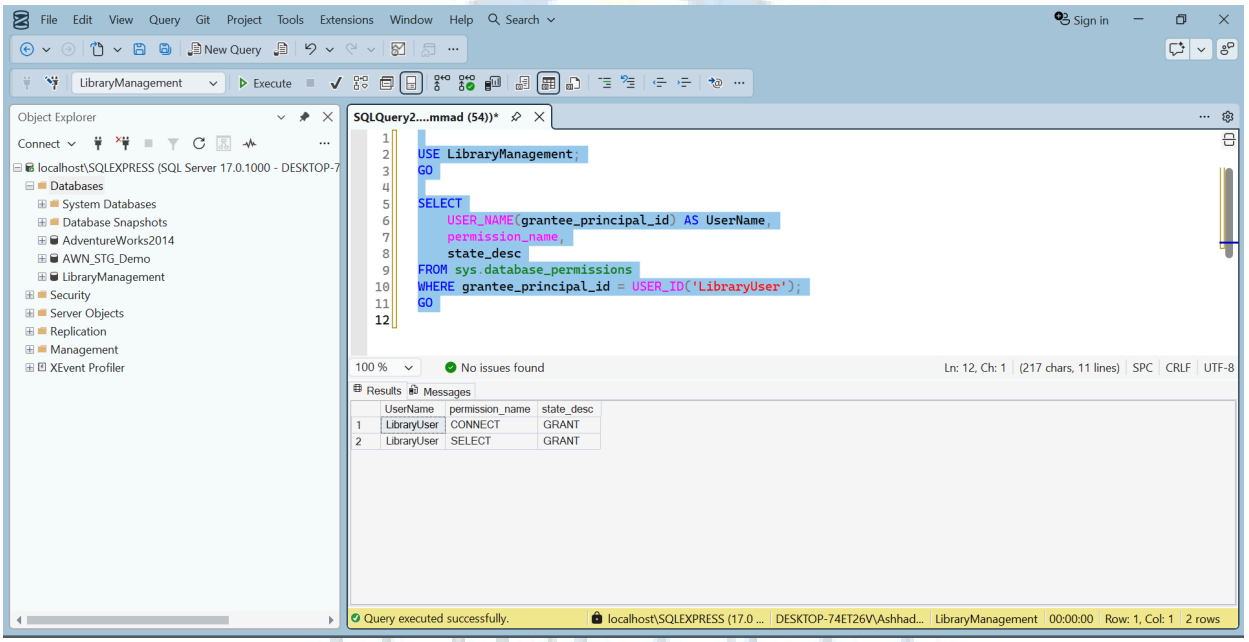
```
REVOKE INSERT ON dbo.Members FROM LibraryUser;

GO
```



Code:

```
USE LibraryManagement;
GO
SELECT
    USER_NAME(grantee_principal_id) AS UserName,
    permission_name,
    state_desc
FROM sys.database_permissions
WHERE grantee_principal_id = USER_ID('LibraryUser');
GO
```



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